# User Manual - Habit Tracking Application

## 1. Introduction

The Habit Tracking Application is a Python-based console system designed to help users monitor daily and weekly habits. It supports predefined habits such as Water Intake, Exercise, Sleep, Reading, and Meditation.

This report presents the completed Habit Tracking Application developed across three phases: conception, implementation, and finalization. The system is designed to help users track daily habits, analyse performance trends, and maintain long-term consistency. The final version integrates a modular architecture, persistent storage via SQLite, and a rich set of analytics accessible through an 11-option command-line menu.

## 2. System Requirements

- Python 3.10 or higher
- SQLite (included automatically with Python)
- Terminal or command-line interface
- Project folder containing all modules and database file

## 3. Starting the Application

1. Open your terminal.
2. Navigate to the project directory:

Code

```
cd habit_tracker
```

3. Run the application:

Code

```
python main.py
```

**4. Main Menu Overview**

When the program starts, you will see a menu similar to:

1. View Habits
2. Record Habit
3. Longest Streak
4. Current Streak
5. Streak Statistics
6. View Records
7. Daily Report
8. Weekly Report
9. Monthly Report
10. Habit Leaderboard
11. Exit

You can select an option by entering its number and pressing Enter.

```
================================
        HABIT TRACKER
================================
1 View Habits
2 Record Habit
3 Longest Streak
4 Current Streak
5 Streak Statistics
6 View Records
7 Daily Report
8 Weekly Report
9 Monthly Report
10 Habit Leaderboard
11 Exit
Select option:
```

*Figure 1: Main menu of the Habit Tracking Application displaying all 11 available options*

## 5. Menu Options in Detail

### 5.1 Option 1 – View Habits

This option displays all predefined habits stored in the system. For each habit, the following information is shown:

- ID
- Name
- Frequency (e.g., daily)

Use this screen to identify the habit ID you need when recording entries or viewing streaks.

```
Select option: 1

ID   Name                      Frequency
---------------------------------------------
1    Water Intake              daily
2    Exercise (Walking)        daily
3    Sleep                     daily
4    Reading                   daily
5    Meditation                daily
```

*Figure 2: Predefined habits with their IDs and daily frequency.*

### 5.2 Option 2 – Record Habit

This option allows you to record a new entry for a selected habit.

**Typical flow:**

1. Enter the habit ID (as shown in “View Habits”).
2. If multiple units are available, select the desired unit (e.g., liters, gallons, hours, minutes).
3. Enter the value (e.g., 2 liters, 30 minutes).
4. The system validates the input and confirms that the entry has been recorded successfully.

Entries are stored in the SQLite database with habit ID, value, unit, and date.

```
Enter habit ID: 1
Available units: gallons, liters
Choose unit: gallons
Enter value (gallons): 2
Recorded: Water Intake = 2.0 gallons on 2026-03-17
```

*Figure 3: Recording a Water Intake entry with unit selection and confirmation.*

```
ID   Name                      Frequency
---------------------------------------------
1    Water Intake              daily
2    Exercise (Walking)        daily
3    Sleep                     daily
4    Reading                   daily
5    Meditation                daily

Enter habit ID: 5
Available units: minutes, hours
Choose unit: hours
Enter value (hours): 6
Recorded: Meditation = 6.0 hours on 2026-04-08
```

*Figure 4: Successful recording of a Meditation habit entry.*

### 5.3 Option 3 – Longest Streak

This option shows the **longest streak** for a specific habit.

**Steps:**

1. Enter the habit ID.
2. The system calculates and displays the maximum number of consecutive days the habit was completed without interruption.

This helps you understand your best historical performance for that habit.

```
Habit ID: 1
Longest streak: 2
```

*Figure 5: Longest streak result for Habit ID 1.*

### 5.4 Option 4 – Current Streak

This option shows the **current streak** for a specific habit.

**Steps:**

1. Enter the habit ID.

2. The system calculates how many consecutive days you have maintained the habit up to today.

This is useful for tracking ongoing consistency.

```
Habit ID: 1
Current streak: 1
```

*Figure 7: Current streak result for Habit ID 1.*

```
7 Daily Report
8 Weekly Report
9 Monthly Report
10 Habit Leaderboard
11 Exit
Select option: 4

ID   Name                      Frequency
---------------------------------------------
1    Water Intake              daily
2    Exercise (Walking)        daily
3    Sleep                     daily
4    Reading                   daily
5    Meditation                daily
Habit ID: 5
Current streak: 1
```

*Figure 8: Current streak displayed through the main menu.*

### 5.5 Option 5 – Streak Statistics

This option provides detailed streak statistics for a habit, such as:

- Maximum streak
- Minimum streak
- Average streak length

**Steps:**

1. Enter the habit ID.
2. The system analyses all recorded streaks and displays the aggregated statistics.

This gives a deeper view of long-term performance patterns.

```
Habit ID: 1
Max streak: 2
Min streak: 1
Average streak: 1.2

-----------------------------
```

*Figure 9: Detailed streak statistics for Habit ID 1.*

```
1    Water Intake              daily
2    Exercise (Walking)        daily
3    Sleep                     daily
4    Reading                   daily
5    Meditation                daily
Habit ID: 5
Max streak: 3
Min streak: 1
Average streak: 1.5

===============================
```

*Figure 10: Performance summary for a selected habit.*

### 5.6 Option 6 – View Records

This option lists all recorded entries for all habits.

Each record typically includes:

- Habit name
- Value
- Unit
- Date

Use this view to inspect raw data and verify that entries have been stored correctly.

```
Select option: 6

Habit Records
Water Intake 2.0 gallons 2026-03-16
Water Intake 4.0 liters 2026-03-16
Water Intake 3.0 liters 2026-03-16
Water Intake 3.0 gallons 2026-03-16
Exercise (Walking) 7.0 km 2026-03-16
Exercise (Walking) 2.0 km 2026-03-16
Exercise (Walking) 4.0 miles 2026-03-16
Sleep 6.0 hours 2026-03-16
Sleep 60.0 minutes 2026-03-16
Sleep 4.0 hours 2026-03-16
Sleep 45.0 minutes 2026-03-16
Reading 56.0 minutes 2026-03-16
Reading 2.0 hours 2026-03-16
Meditation 35.0 minutes 2026-03-16
Meditation 1.0 hours 2026-03-16
Meditation 1.5 hours 2026-03-16
Water Intake 2.0 gallons 2026-03-17
Water Intake 4.0 liters 2026-03-17
Exercise (Walking) 7.0 km 2026-03-17
Exercise (Walking) 4.0 miles 2026-03-17
Sleep 5.0 hours 2026-03-17
```

*Figure 11: Complete list of recorded habit entries with values, units, and timestamps.*

### 5.7 Option 7 – Daily Report

This option generates a **daily report** for the current day.

The report shows, for each habit:

- Whether it was completed today (e.g., “Completed” / “Not completed”)

This provides a quick overview of your performance for the current day.

```
Daily Report
-------------
Water Intake     Completed
Exercise (Walking) Completed
Sleep            Completed
Reading          Completed
Meditation       Completed

--------------------------------
```

*Figure 12: Daily report showing completion status for all habits.*

### 5.8 Option 8 – Weekly Report

This option generates a **weekly report** summarizing the last 7 days.

For each habit, the report may show:

- Number of days completed out of 7
- A progress indicator or percentage

This helps you evaluate short-term consistency across the week.

```
Weekly Report
-------------
Water Intake      28% (2/7)
Exercise (Walking)  28% (2/7)
Sleep             28% (2/7)
Reading           28% (2/7)
Meditation        28% (2/7)
```

*Figure 13: Weekly report summarizing habit completion across the last 7 days.*

### 5.9 Option 9 – Monthly Report

This option generates a **monthly report** summarizing the last 30 days.

For each habit, the report may show:

- Number of days completed out of 30
- Completion percentage
- Visual progress representation (e.g., progress bars)

This supports long-term trend analysis.

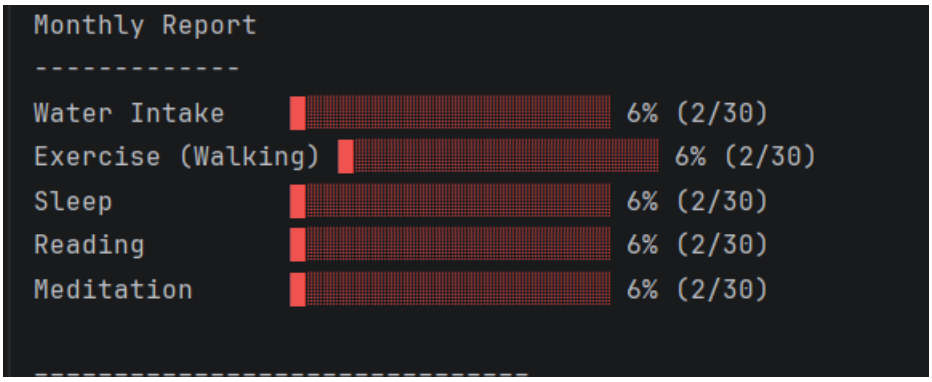


*Figure 14: Monthly report summarizing habit performance over 30 days.*

### 5.10 Option 10 – Habit Leaderboard

This option displays a **leaderboard** ranking habits by performance.

Typical information includes:

- Habit ranking
- Completion percentage or score per habit

The leaderboard highlights which habits are most consistently maintained.

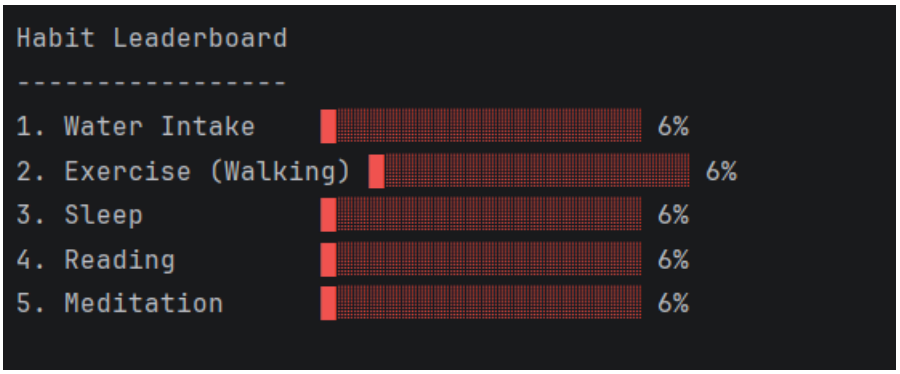


*Figure 15: Habit leaderboard ranking habits by consistency*

### 5.11 Option 11 – Exit

This option closes the application.

When you select 11, the program terminates safely and returns you to the terminal.

### 6. Data Storage

All habit entries are stored in a local SQLite database (e.g., data/habits.db). The database ensures:

- Persistent storage across sessions
- Efficient querying for analytics
- Reliable streak calculations

### 7. Troubleshooting

- **Database not found** → Ensure the data folder exists.
- **Invalid input** → Enter numeric values where required.
- **Habit not updating** → Check if the habit allows multiple entries per day.
- **No habits shown:** Ensure the database and predefined habit setup are correctly initialized.
- **Invalid input errors:** Use numeric values where required and valid habit IDs.
- **No streaks or reports:** Make sure you have recorded entries for the relevant period.

### 8. Conclusion.

The Habit Tracking Application provides a structured way to record habits, analyze performance, and maintain consistency over time. The 11-option menu offers fine-grained access to streak analytics, detailed records, and summary reports.

### APPENDIX — TECHNICAL DETAILS

```
habits = {
    1: {"name": "Water Intake", "frequency": "daily", "unit": "gallons", "options": ["gallons
    2: {"name": "Exercise (Walking)", "frequency": "daily", "unit": "km", "options": ["km", '
    3: {"name": "Sleep", "frequency": "daily", "unit": "hours", "options": ["hours", "minutes
    4: {"name": "Reading", "frequency": "daily", "unit": "hours", "options": ["hours", "minut
    5: {"name": "Meditation", "frequency": "daily", "unit": "minutes", "options": ["minutes",
}
```

*Figure 16: Python dictionary defining the five predefined habits.*

```
Run   main   habit_tracker_demo

C:\Users\mista\PycharmProjects\PythonProject\.venv\Scripts\python.exe C:\Users\mista\PycharmProjects\PythonProject\tests\habit_tracker_demo.py
Test Case 1
Habit: Study Python
Date: 2026-03-01
Output: Habit recorded successfully.
--------------------------
Test Case 2
Habit: Exercise
Date: 2026-03-02
Output: Habit recorded successfully.
--------------------------

Process finished with exit code 0
```

*Figure 17: Automated test script confirming successful habit recording.*

```
Habit Records
Water Intake 2.0 gallons 2026-03-16
Water Intake 4.0 liters 2026-03-16
Water Intake 3.0 liters 2026-03-16
Water Intake 3.0 gallons 2026-03-16
Exercise (Walking) 7.0 km 2026-03-16
Exercise (Walking) 2.0 km 2026-03-16
Exercise (Walking) 4.0 miles 2026-03-16
Sleep 6.0 hours 2026-03-16
Sleep 60.0 minutes 2026-03-16
Sleep 4.0 hours 2026-03-16
Sleep 45.0 minutes 2026-03-16
Reading 56.0 minutes 2026-03-16
Reading 2.0 hours 2026-03-16
Meditation 35.0 minutes 2026-03-16
Meditation 1.0 hours 2026-03-16
Meditation 1.5 hours 2026-03-16
Water Intake 2.0 gallons 2026-03-17
Water Intake 4.0 liters 2026-03-17
Exercise (Walking) 7.0 km 2026-03-17
Exercise (Walking) 4.0 miles 2026-03-17
Sleep 5.0 hours 2026-03-17
Reading 2.0 hours 2026-03-17
Meditation 75.0 minutes 2026-03-17
Water Intake 2.0 liters 2026-03-18
Exercise (Walking) 5.0 miles 2026-03-18
Sleep 6.0 hours 2026-03-18
```

*Figure 18: Generating dummy data for testing the habit tracker.*